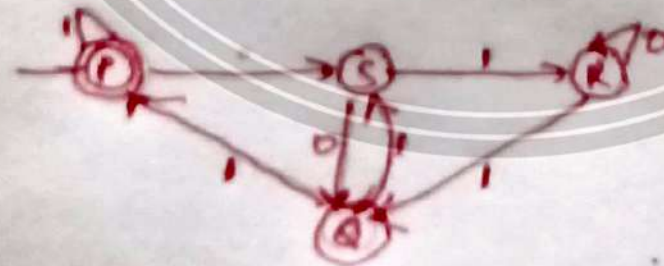


UNIT-I QUESTION BANK

1. Obtain DFAs to accept strings of a's and b's having exactly one a.
2. Obtain a DFA to accept strings of a's and b's having even number of a's and b's.
3. Give Applications of Finite Automata.
4. Define DFA, NFA & Language?
5. Obtain a DFA to accept strings of a's and b's starting with the string ab.
6. Draw a DFA to accept string of 0's and 1's ending with the string 011.
7. Write DFA to accept strings of 0's, 1's & 2's beginning with a 0 followed by odd number of 1's and ending with a 2.
8. Design a DFA to accept string of 0's & 1's when interpreted as binary numbers would be multiple of 3.
9. Find closure of each state and give the set of all strings of length 3 or less accepted by automaton.
10. Obtain a DFA to accept strings of a's and b's having a sub string aa
11. Write Regular expression for the following $L = \{ a^n b^m : m, n \text{ are even} \}$ $L = \{ a^n, b^m : m \geq 2, n \geq 2 \}$.
12. Convert following NFA to DFA

δN	0	1
p	{p,r}	{q}
q	{r,s}	{p}
*r	{p,s}	{r}
*s	{q,r}	

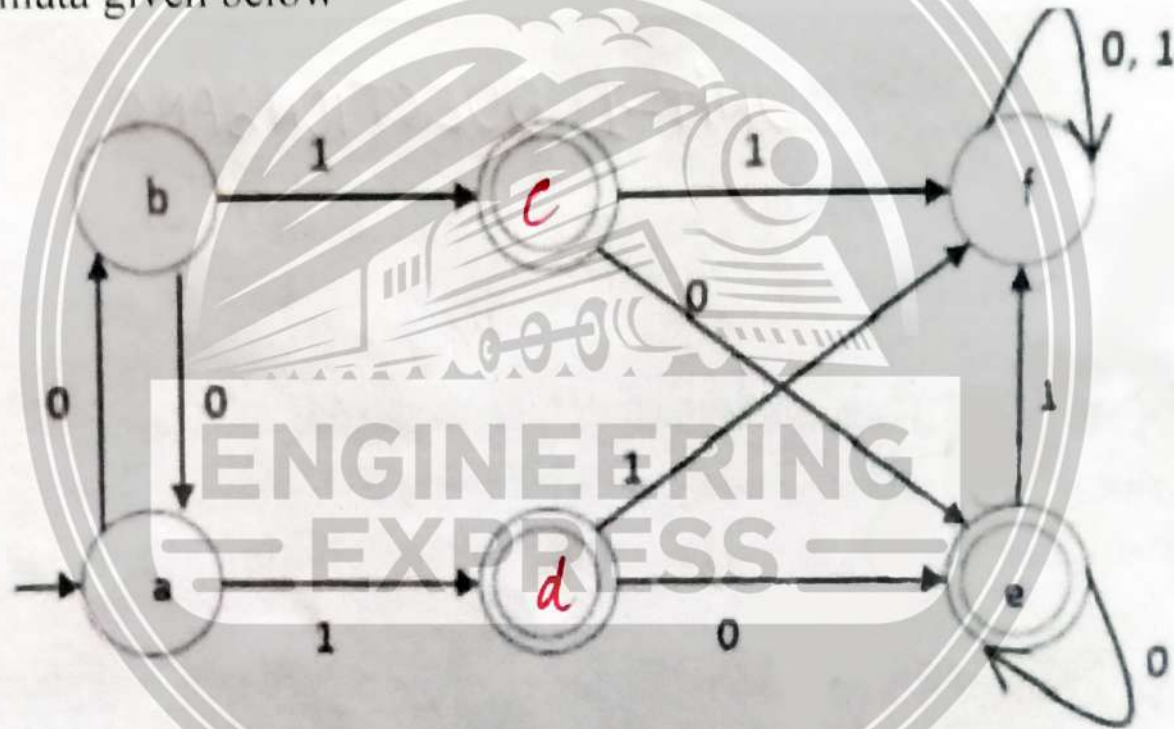
13. Convert the following DFA to Regular Expression



14. Define NFA. With example explain the extended transition function.

15. Explain the ground rules of finite automata.

16. Minimize the automata given below

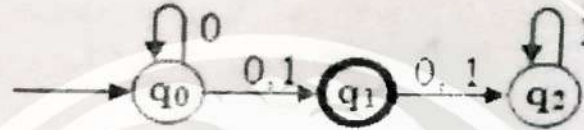


17. State the Myhill Merode Theorem. Write the steps of Myhill Merode Theorem when applying to minimization a DFA.

18. Define grammar? Explain Chomsky Hierarchy? Give an example

UNIT-II QUESTION BANK

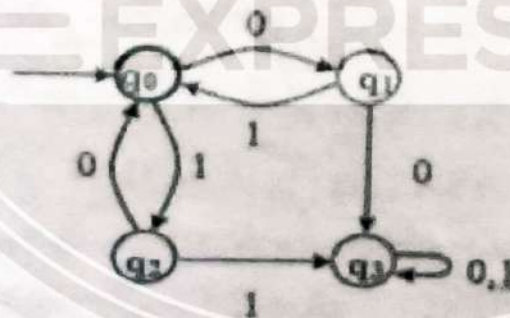
1. Define Regular Grammar and Regular Language with Example
2. What is Regular Expression write its rule.
3. Obtain an NFA to accept the following language $L = \{w | wabab^n \text{ or } aba^n \text{ where } n \geq 0\}$
4. Convert the following NFA into an equivalent DFA



5. Obtain an NFA which accepts strings of a's and b's starting with the string ab
6. Prove pumping lemma?
7. Prove that $L = \{w | w \text{ is a palindrome on } \{a,b\}^*\}$ is not regular. i.e., $L = \{aabaa, aba, abbbba, \dots\}$
8. Prove that $L = \{\text{all strings of 1's whose length is prime}\}$ is not regular. i.e., $L = \{1^2, 1^3, 1^5, 1^7, 1^{11}, \dots\}$
9. What is the language accepted by the following FA



10. Write short note on Applications of Regular Expressions.
11. Obtain a regular expression for the FA shown below



12. Explain Closure properties with an example
13. State and Proof Kleen,s Theorem with example.
14. Construct the FA for $RE := (0+1)^*(00+11)(0+1)^*$
15. State and Proof Ardens Theorem.
16. What is Moore and Mealy Machine.

UNIT-III QUESTION BANK

1. Define context free grammar and context free language with examples.
2. Define derivation, types of derivation, Derivation tree & ambiguous grammar. Give example for each.
3. Is the following grammar ambiguous?
 $S \rightarrow aB \mid bA$
 $A \rightarrow aS \mid bAA \mid a$
4. Obtain grammar to generate string consisting of any number of a's and b's with at least one b.
5. Obtain a grammar to generate the following language $L = \{WW^R \mid W \in \{a, b\}^*\}$.
6. Obtain a grammar to generate the following language: $L = \{0^m 1^m 2^n \mid m \geq 1 \text{ and } n \geq 0\}$
7. Convert the following CFG into CNF
 $S \rightarrow XY \mid Xn \mid p$
 $X \rightarrow mX \mid m$
 $Y \rightarrow Xn \mid o$
8. Convert the following CFG into CNF: $S \rightarrow ASA \mid aB, A \rightarrow B \mid S, B \rightarrow b \mid \epsilon$
9. Consider the grammar with the production $S \rightarrow aSS, A \rightarrow b$. Compute the string aababbb with the left most and right most derivation. Draw the derivation tree.
10. Find a reduced grammar equivalent to the grammar G, having production rules, $P: S \rightarrow AC \mid B, A \rightarrow a, C \rightarrow c \mid BC, E \rightarrow aA \mid e$
11. Remove unit production from the following $S \rightarrow XY, X \rightarrow a, Y \rightarrow Z \mid b, Z \rightarrow M, M \rightarrow N, N \rightarrow a$
12. Remove null production from the following $S \rightarrow ASA \mid aB \mid b, A \rightarrow B, B \rightarrow b \mid \epsilon$
13. Convert the following grammar G into Greibach Normal Form (GNF) $S \rightarrow XA \mid BB, B \rightarrow b \mid SB, X \rightarrow b, A \rightarrow a$
14. The language $L = \{a^n b^n c^n \mid n \geq 1\}$ is not context free. Proof.
15. Find out whether the language $L = \{x^n \mid n \geq 0\}$ is context free or not.

UNIT -IV

1. Explain push down automata with an example
2. Construct PDA: For the language

$$L = \{wcw^R \mid w \in \{a, b\}^*, c \in \Sigma\}$$

3. Construct a PDA that accepts $L = \{ww^R \mid w = (a+b)^*\}$
4. Construct PDA to accept $L = \{0^n 1^n \mid n > 0\}$
5. Describe **Instantaneous Description** (ID) of PDA.
6. Differentiate PDA by Empty Stack and Final State by giving their definition
7. Define various type of PDA.
8. Construct NPDA for $L = \{a^n b^n c^{n+m} \mid m, n \geq 0\}$
9. Construct a PDA from the following CFG. $G = (\{S, X\}, \{a, b\}, P, S)$ where the productions are –
 $S \rightarrow XS \mid \epsilon, A \rightarrow aXb \mid Ab \mid ab$
10. Construct a PDA from the following CFG
 $S \rightarrow aABB/aAA, A \rightarrow aBB/a, B \rightarrow bBB/A$
11. Construct CFG from the following PDA

PDA $M = (\{q_0\}, \{0,1\}, \{a,b,z_0\}, \delta, q_0, z_0, \phi)$

δ is defined as

$$\delta(q_0, a, z_0) = \{(q_0, az_0)\}$$

$$\delta(q_0, 1, z_0) = \{(q_0, bz_0)\}$$

$$\delta(q_0, 0, a) = \{(q_0, aa)\}$$

$$\delta(q_0, 1, b) = \{(q_0, bb)\}$$

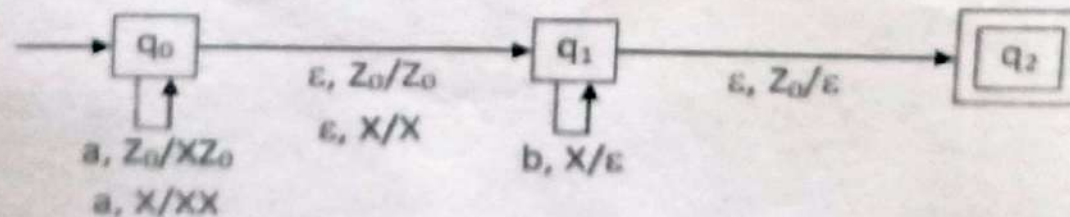
$$\delta(q_0, 0, b) = \{(q_0, \epsilon)\}$$

$$\delta(q_0, 1, a) = \{(q_0, \epsilon)\}$$

$$\delta(q_0, \epsilon, z_0) = \{(q_0, \epsilon)\}$$

Convert given PDA M to corresponding CFG

12. What is two stack PDA. Design Two Stack PDA for $L = \{a^n b^n c^n \mid n \geq 1\}$
13. Converting a PDA into an equivalent CFG



UNIT - V

1. What is Turing Machine? Describe the various methods to represent it.
2. Post's Correspondence problem Design a TM to recognize a string of 0s and 1s such that the number of 0s is not twice as that of 1s
3. Describe the Language Generated by Turing Machine.
4. Design a TM to recognize all strings consisting of an odd number of a 's.
5. Prove that the halting problem is undecidable.
6. Prove that single tape machines can simulate multi-tape machines.
7. Design a TM to recognize $L = \{a^n b^n \mid n \geq 1\}$
8. Design a TM to recognize $L = \{a^{n+2} b^n \mid n \geq 0\}$
9. Design a Turing machine to accept a Palindrome
 $L = \{ww^r \mid w \in (a,b)^*\}$
 $L = \{waw^r \mid w \in (a,b)^*\}$
10. Design a TM to recognize a string of the form $a^n b^{2n}$
11. Design a TM to recognize $L = \{a^n b^n c^n \mid n \geq 1\}$
12. Define undecidability, decidability
13. Write Short Notes on
 - a. Church- Thesis
 - b. Multitape TM
 - c. Recursive Enumerable Languages
14. Design a TM compute Addition and subtraction of two Unary number.